



Cambridge International AS & A Level

CANDIDATE
NAME



CENTRE
NUMBER

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FURTHER MATHEMATICS

9231/31

Paper 3 Further Mechanics

October/November 2024

1 hour 30 minutes

You must answer on the question paper.

You will need: List of formulae (MF19)

INSTRUCTIONS

- Answer **all** questions.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen or correction fluid.
- Do **not** write on any bar codes.
- If additional space is needed, you should use the lined page at the end of this booklet; the question number or numbers must be clearly shown.
- You should use a calculator where appropriate.
- You must show all necessary working clearly; no marks will be given for unsupported answers from a calculator.
- Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place for angles in degrees, unless a different level of accuracy is specified in the question.
- Where a numerical value for the acceleration due to gravity (g) is needed, use 10 ms^{-2} .

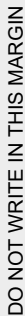
INFORMATION

- The total mark for this paper is 50.
- The number of marks for each question or part question is shown in brackets [].

This document has **16** pages. Any blank pages are indicated.

- Find, in either order, the value of u and the value of H . [5]

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- [5]

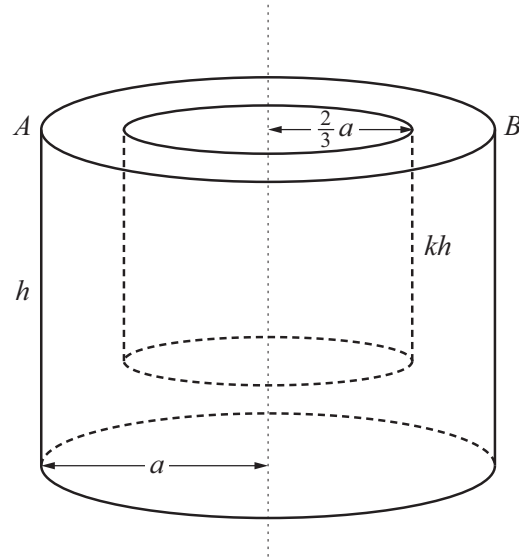
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3 A particle P of mass m kg is attached to one end of a light elastic string of natural length 2 m and modulus of elasticity $2mg$ N. The other end of the string is attached to a fixed point O . The particle P hangs in equilibrium vertically below O . The particle P is pulled down vertically a distance d m below its equilibrium position and released from rest.

(a) Given that the particle just reaches O in the subsequent motion, find the value of d . [6]

[illegible]



An object is formed by removing a cylinder of radius $\frac{2}{3}a$ and height kh ($k < 1$) from a uniform solid cylinder of radius a and height h . The vertical axes of symmetry of the two cylinders coincide. The upper faces of the two cylinders are in the same plane as each other. The points A and B are the opposite ends of a diameter of the upper face of the object (see diagram).

- (a) Find, in terms of h and k , the distance of the centre of mass of the object from AB . [4]

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(b) Find the magnitude of F when $t = 3$.

[3]

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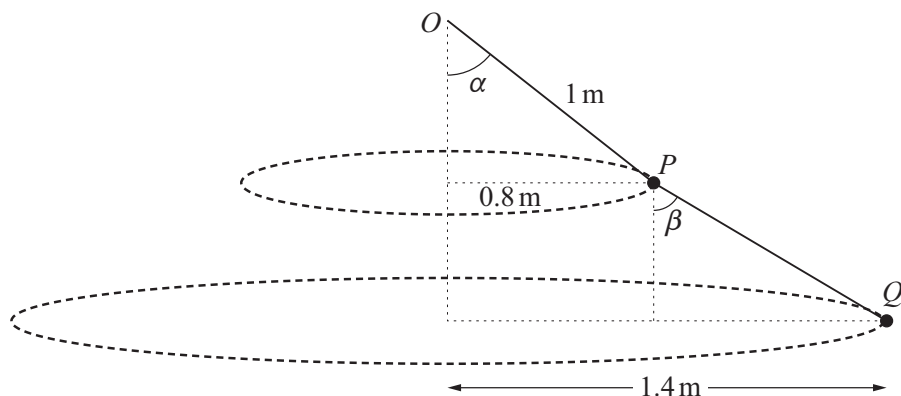
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A particle P of mass 0.05 kg is attached to one end of a light inextensible string of length 1 m . The other end of the string is attached to a fixed point O . A particle Q of mass 0.04 kg is attached to one end of a second light inextensible string. The other end of this string is attached to P .

The particle P moves in a horizontal circle of radius 0.8 m with angular speed $\omega\text{ rad s}^{-1}$. The particle Q moves in a horizontal circle of radius 1.4 m also with angular speed $\omega\text{ rad s}^{-1}$. The centres of the circles are vertically below O , and O , P and Q are always in the same vertical plane. The strings OP and PQ remain at constant angles α and β respectively to the vertical (see diagram).

- (a) Find the tension in the string OP . [3]

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[3]

[illegible]

[2]

[illegible]

- [4]

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(b) Find the value of e . [6]

[illegible]

[illegible]

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